

PRISM Platform — Complete Patient & Admin Portal Guide (Sprint 3.5)

Audience: Clinical operators, product owners, and non-technical stakeholders

Language: Plain English (layperson-friendly)

Version: PRISM 2.0 · Black & Pink One · Sprint 3.5

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1. What Is PRISM?

PRISM stands for **Patient-centric Retrieval Intelligence System for Medicine**.

In simple terms, PRISM is a **healthcare AI platform** with two faces:

Portal	Who uses it	Purpose in one sentence
Patient Portal	People seeking health guidance	Chat with specialist AI “companions” that answer using verified medical documents , not guesses.
Admin Portal (Command Centre)	Medical directors, operators, admins	Watch system health, measure AI quality, manage documents, and see when patients need human help.

The core idea: RAG (why answers are safer)

Most chatbots “make things up” when they do not know an answer. That is dangerous in medicine.

PRISM uses **RAG (Retrieval-Augmented Generation)**:

1. You ask a question.
2. The system **searches its medical library** (PubMed, CDC, uploaded guidelines, etc.).
3. It gives those facts to the AI and says: *“Answer only from these paragraphs.”*
4. You get a reply with **sources (citations)** where possible.

Think of it as an **open-book exam** for the AI, not a memory test.

Clinical structure: 5 diseases × 30 agents

Disease domain	Code	What it covers (plain language)
Cancer Care	CA	Screening, treatment paths, support, survivorship, genetics
Diabetes	DM	Glucose, insulin, nutrition, complications, pregnancy/special cases
Cardiovascular	CV	Heart risk, emergencies, meds, rehab, diet
Mental Illness	MH	Depression, anxiety, sleep, trauma, crisis support
Respiratory	RS	Asthma, COPD, rehab, medications, sleep apnea

Each domain has **6 AI agents** (30 total) — like a small team of virtual specialists, each focused on one slice of care (e.g. “Diabetes Nutrition Specialist” vs “Insulin Management Specialist”).

2. How the Two Portals Fit Together

Diagram (see online Markdown viewer for full graphic):

```
flowchart LR
    subgraph public [Public entry]
        Home["Home page /"]
        PLand["Patient marketing /patient"]
    end
    subgraph patient [Patient Portal]
        LoginP["Login patient"]
        Dash["Dashboard..."]
    end
```

- Patient journey (typical):**
Home → Patient landing → Sign in → Dashboard (subscriptions overview) → **Chat app** (main experience).
- Admin journey (typical):**
Home → Admin login → **Admin intro** (capabilities overview) → **Command Centre** (17 sidebar modules).

3. Technology Stack (Tools Behind the Scenes)

3.1 Frontend (what you see in the browser)

Tool	Role	Layman explanation
React 18	UI framework	Building blocks for screens, buttons, chat bubbles
Vite 5	Build & dev server	Fast local website; proxies API calls to backend
React Router 6	Page navigation	Moves you between /app, /admin, /login, etc.
Tailwind CSS 3	Styling	Consistent colors, spacing, dark themes
Zustand 5	App state	Remembers who is logged in (token in browser storage)
Axios	HTTP client	Sends messages and loads admin charts from the API
Recharts 2	Charts	Line/bar/pie graphs in admin dashboards
Lucide React	Icons	Sidebar and button icons
React Markdown	Rich text	Formats AI answers with lists and tables
React Dropzone	File upload	Drag-and-drop medical PDFs in admin
React Hot Toast	Notifications	“Success” / “Error” popups
Radix UI	Accessible widgets	Dialogs, tabs, tooltips
date-fns	Dates	“Last 7 days” style labels

Two frontend run modes:

Mode	Command	URL	Storage key
Patient	npm run dev	http://localhost:5177	prism-auth

Admin	<code>npm run admin</code>	<code>http://localhost:5178</code>	<code>prism-auth-admin</code>
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3.2 Backend (the “engine room”)

Tool	Role	Layman explanation
Python 3.11+	Language	All server logic
FastAPI	Web API	REST endpoints (<code>/api/chat</code> , <code>/api/admin/...</code>)
Uvicorn	Server	Runs the API on port 8000
SQLAlchemy 2 (async)	Database ORM	Talks to PostgreSQL
asyncpg	DB driver	Fast connection to Postgres
PostgreSQL 16	Main database	Users, chats, feedback, alerts, metrics
Redis 7	Cache (optional)	Speed layer when configured
ChromaDB	Vector database	Stores “chunks” of medical text for search
LangChain / LangGraph	AI orchestration	Chains retrieval + LLM steps
OpenAI / Anthropic APIs	Large language models	Generates patient-facing answers
sentence-transformers	Embeddings	Turns text into searchable vectors
RAGAS	Quality scoring	Automated grades for AI answers
JWT (python-jose)	Security tokens	Proves you are logged in as patient or admin
bcrypt	Password hashing	Stores passwords safely
deep-translator / langdetect	Multilingual	Detects language and translates
Whisper (OpenAI)	Speech-to-text	Voice messages
Pillow / Tesseract	Images & OCR	Reads text from uploaded images
BeautifulSoup / Biopython	Crawlers	Fetches PubMed & CDC content

fpdf2 / pdfplumber	PDFs	Prescriptions and history downloads
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3.3 Infrastructure (Docker)

Service	Port	Purpose
postgres	5432	User accounts, conversations, metrics
redis	6379	Caching (when enabled)
chromadb	8001	Remote vector store (falls back to local <code>./data/chroma_db</code> if offline)

Start databases:

```
docker compose up -d postgres redis chromadb python scripts/setup_db.py python run.py
```

4. Getting In — URLs, Accounts & Security

4.1 Main URLs (local development)

Page	URL
Platform home	http://localhost:5177/
Patient marketing	http://localhost:5177/patient
Patient login	http://localhost:5177/login
Patient dashboard	http://localhost:5177/dashboard
Patient chat (main app)	http://localhost:5177/app
Admin login	http://localhost:5178/login?role=admin
Admin welcome	http://localhost:5178/admin-intro
Admin Command Centre	http://localhost:5178/admin
API docs (Swagger)	http://localhost:8000/docs

4.2 Demo accounts

Role	Email	Password
Admin	admin@prism.ai	admin123
Patient	patient@prism.ai	demo123
Patient (alt)	demo@prism.ai	demo123

4.3 Security (how access is controlled)

- **JWT tokens:** After login, the browser stores a token. Every API call sends `Authorization: Bearer <token>`.
- **Role enforcement:** Admin routes require `role: admin` in the token. Patient login at `/login?role=admin` rejects non-admin users with “*Authorized admin access only.*”
- **Protected routes:** `/app`, `/dashboard`, `/admin`, `/admin-intro` redirect to login if not signed in.
- **Important:** If **PostgreSQL is stopped**, the backend may use a limited “development login” that does not load real admin data. Always run Postgres for production-like admin use.

5. Patient Portal — Full Feature Guide

5.1 Public & marketing pages

Home (`/`)

- **Dual portal launcher:** Choose Patient Portal or Admin Console.
- **Theme switcher:** Black & Pink, Titanium & Gold, Neural Glass.
- **Device preview:** Desktop / tablet / mobile frame on the home page.
- **Platform stats:** 5 disease domains, 30 agents, 5 languages, 19-dimension quality gates.

Patient landing (`/patient`)

Marketing page describing:

- 5 disease areas and 30 specialist agents
- Evidence from PubMed, CDC, WHO
- Multilingual, multimodal (text, image, voice)
- Prescription download, chat history, feedback, document upload

Button: **Get Started** → patient login.

5.2 Authentication (`/login`)

Feature	What it does
Sign in	Email + password; returns JWT and user profile
Create account	Name, email, password, LATAM country (patient flow only)
Admin vs patient mode	URL <code>?role=admin</code> shows “Admin Console” branding and blocks non-admin users
Demo credentials	Shown on screen for quick testing
DB offline warning	Toast if database unavailable (limited dev mode)

After login, patients go to `/app`; admins go to `/admin-intro`.

5.3 Patient dashboard (`/dashboard`)

A **home base** before deep chat (subscription-oriented layout):

Section	Layman description
Greeting & date	Personalized welcome using your account name
Stats strip	Active care plans, agent count, conversation volume, billing/trial reminder (demo values)
Active care plans	Cards per subscribed disease (CA, DM, CV, MH, RS) with price, renewal, and 5–6 agent codes
Health snapshot	Sample vitals (glucose, HbA1c, BP, BMI, steps) — illustrative wellness panel
Quick actions	Jump into chat agents; link to Manage subscriptions (<code>/plans</code>)

5.4 Main chat application (`/app`) — the heart of the patient experience

This is the **primary patient product**: a full-screen chat workspace.

A. Disease & agent sidebar

- Lists **5 disease domains** you are subscribed to (demo: all five enabled).
- Under each disease, **6 specialist agents** with icons and short descriptions.

- Click an agent → starts or **resumes** your last conversation with that agent.
- **History button** opens the 15-day chat history panel.

Layman analogy: Like picking which “department” and which “doctor personality” you want to talk to.

B. Agent header bar (when chatting)

Control	Purpose
Agent name & disease	Shows who you are talking to
Routing badge	Shows if you are on Primary, Specialist, or Human escalation path
Confidence indicator	How sure the AI is about its answer (0–100%)
Language selector	Switch UI and chat language
Prescription	Download AI-generated prescription PDF (after conversation exists)
History download	Export current chat as PDF
Chat history panel	Browse past sessions
Logout	Ends session

C. Text chat (core)

Feature	Layman explanation
Type a message	Sends to <code>/api/chat</code> with your agent and language
Suggested questions	Tap chips to ask common starters (loaded per agent)
AI answers with citations	Sources from the medical library when available
Follow-up questions	AI may suggest what to ask next
Clarifying questions	AI can ask up to ~5 structured questions before giving a full answer (“conversational engine”)
Intent detection	Understands if you want info, symptom check, medication help, etc.

Quality hint	Internal conversation quality score may surface recommendations
Auto-resume	On open, restores last active conversation across agents
Per-agent resume	Switching agents reloads that agent's latest thread

D. Smart routing & escalation (safety net)

The backend continuously evaluates:

Trigger	What happens (patient-visible)
Low confidence (< 70%)	Route to Specialist Agent tier (deeper model path)
High frustration (> 75/100)	Human coordinator card appears with contact-style info
Escalation monitor	Tracks emotional and clinical risk signals

Layman explanation: If the AI is unsure or you seem very upset, PRISM escalates to a stronger AI layer or shows how to reach a human — similar to a nurse triage line.

E. Multimodal inputs

Modality	How it works
Image upload	Send lab reports, prescriptions, or medical photos; vision model extracts values and explains in plain language; flags critical values
Voice	Record audio → transcribed → answered via voice chat pipeline; playback supported
Documents	PDF/images processed through multimodal pipeline (admin-ingested content powers RAG; patient uploads analyzed per message)

F. Multilingual support

Supported languages include **English, Spanish, Hindi, Telugu, Punjabi** (and UI strings for each).

- Detects or respects selected language
- Can show **transliteration banner** for non-Latin scripts
- Translates user input and/or responses as configured

G. Chat history (15-day memory)

Opened from sidebar or header **History**:

Capability	Description
Timeline view	Conversations grouped: Today, Yesterday, This week, etc.
Filter by disease / agent	Narrow to one specialty
Topic grouping	Organized by conversation title/topic
Search	Full-text search across past messages (min 2 characters)
Restore session	Click a past chat → reloads messages into the main window
Delete one chat	Removes a conversation and its messages
Clear all history	Wipes your archived sessions
Hide conversation	Toggle visibility (privacy); hidden threads can be excluded from default view

H. Prescription & export

Action	Output
Generate prescription PDF	Summary-style prescription document for current conversation
Download chat history PDF	Printable transcript of the session

Disclaimer: These are AI-assisted informational exports, not legal prescriptions unless reviewed by a licensed clinician.

I. Patient feedback (after each AI reply)

- **Star rating (1–5)** on assistant messages
- **5 stars** → quick thank-you
- **Below 5** → modal with tags (incorrect, slow, safety concern, etc.) and optional comment
- Feeds **Admin** → **Patient Feedback** and **Quality Score** dashboards

J. Themes & layout

- Uses global **theme** from home (CSS variables: accent color, backgrounds)
- **Portal layout** adds top navigation on non-chat pages; chat is full-height
- **Device switcher** on home simulates mobile/tablet width

5.5 Subscription pages (supporting)

Routes exist for `/plans`, `/checkout`, `/subscribe` (subscription tier selection tied to backend `/api/subscribe`). The dashboard links to plan management — tiers include **Free (1 disease)**, **Basic (2)**, **Premium (5)**.

5.6 Patient portal — backend APIs used

Area	Endpoints (examples)
Auth	POST <code>/api/auth/token</code> , POST <code>/api/auth/register</code> , GET <code>/api/auth/me</code>
Chat	POST <code>/api/chat</code> , POST <code>/api/chat/image</code> , POST <code>/api/voice/chat</code>
Conversations	GET <code>/api/conversations/last_active</code> , GET <code>/api/conversations/resume/{agent_id}</code>
History	GET <code>/api/history</code> , GET <code>/api/history/search</code> , DELETE <code>/api/history/{id}</code>
Export	POST <code>/api/chat/prescription</code> , POST <code>/api/chat/history/download</code>
Feedback	POST <code>/api/feedback</code>
Agents	GET <code>/api/diseases</code> , GET <code>/api/agents/{id}/questions</code>

6. Admin Portal (Command Centre) — Full Feature Guide

The **Command Centre** is the executive cockpit at `/admin`, preceded by a welcome screen at `/admin-intro`.

6.1 Admin intro (`/admin-intro`)

A **guided landing** before the full dashboard:

- Cards for RAGAS, Governance, Pre-RAG, Feedback, Upload & Crawl, Agent Performance, Alerts, Health, Escalations, Quality

- **Enter Command Centre** button
- Footer: “Authorized Access Only • Live Data Pipeline”

6.2 Command Centre layout

UI area	Description
Left sidebar	17 modules (see below)
Top header	Link back to Patient Portal, module title, Force Refresh (reloads all admin APIs)
Main panel	Charts, tables, and controls for the active module
Sign out	Clears admin token

Data loads on entry via parallel API calls (overview, RAGAS, documents, feedback, alerts, vector store, LLM calls, query sources, sentiment). Partial failures still show what loaded.

6.3 Admin modules (sidebar) — detailed

1. System Overview

Layman: The “heartbeat monitor” of the whole platform.

Element	Meaning
Total patients	Registered user count
LLM interactions	How many AI replies were sent
Active sessions	Recent conversation activity
RAGAS composite	Overall AI quality score (0–100%)
Infrastructure health	Live ping status for API, ChromaDB, LLM provider, Pre-RAG gate
Performance trend chart	System availability over time of day
LLM call telemetry	Recent model usage
User query sources	Where patient questions originated

APIs: GET /api/admin/overview, GET /api/admin/health, GET /api/admin/llm-calls, GET /api/admin/user-query-sources

2. RAGAS Intelligence Metrics

Layman: The automated “grading room” for AI answers.

Six dimension tabs:

1. **Retrieval quality** — Did we find the right documents?
2. **Generation quality** — Was the answer faithful and relevant?
3. **Context efficiency** — Was the context concise and useful?
4. **End-to-end accuracy** — Combined score
5. **Safety & harm** — Disclaimers, refusals, harmless wording
6. **NLP benchmarks** — BERT, BLEU, ROUGE, METEOR, perplexity

Includes **trend chart** and **recent evaluations table** (by disease, agent, scores, timestamp).

API: GET /api/admin/ragas

3. Responsible AI Scorecard

Layman: Regional fairness and compliance dashboard for **Latin America**.

- Select **country** (Brazil, Mexico, Colombia, etc.) and **specialty**
- Scores for fairness, cultural safety, health equity, data privacy (LGPD-style), language access, transparency, digital inclusion
- **Recalculate** runs compliance checks

Built for territorial governance reviews.

4. Pre-RAG Readiness

Layman: “Border control” for documents before they enter the AI library.

Tier	Max points	Checks
Tier 1 — Guardrails	40	OCR quality, duplicates, copyright, freshness, PDF layout, PII, offensive content, metadata
Tier 2 — Evidence	60	Source authority, evidence grade, peer review, recency, LATAM relevance, clinical focus, etc.

Readiness tiers: GOLD (≥85), SILVER (70–84), BORDERLINE (55–69), REJECTED (<55)

- Table of all ingested files with scores
- **View Report** → 19-dimension breakdown modal

API: GET /api/admin/prerag/report

5. Vector Store

Layman: The map of the **medical knowledge vault** (ChromaDB collections).

- ~**30 collections** (5 diseases × 6 agent libraries)
- Chunk counts and memory bars per collection
- Shows **Remote** vs **Local fallback** connection mode

API: GET /api/admin/vector-store

6. Patient Feedback

Layman: Satisfaction HQ — what patients rated after chats.

- Average rating and **star distribution** (5★ down to 1★)
- Filter by specialty
- **Critical feedback drawer** — comments below 5★
- Full feedback table: agent, disease, rating, tags, comment, date

API: GET /api/admin/feedback

7. Sentiment Analysis

Layman: Emotional weather map — how patients *feel* during care.

View	Content
Overall stats	Distress vs equanimity vs reassured counts
Disease-wise bar chart	Average sentiment by therapeutic area
Taxonomy pie	Emotional categories
Patient directory	Sentiment category, stars, frustration %, last active
Patient drawer	Frustration trend line, clinical action plan, cognitive alerts

API: GET /api/admin/sentiment

8. Indexed Documents

Layman: Card catalog of every document in the AI library.

Columns: title, assigned agent, specialty, source, chunk count, Pre-RAG badge (GOLD/SILVER/BORDERLINE), ingest date.

API: GET /api/admin/documents

9. Upload & Crawl

Layman: Import terminal for new medical knowledge.

Side	Function
Upload	Drag PDF, TXT, DOCX, images → validation pipeline (medical F1, disease match, agent match, vector readiness)
Crawl	Search PubMed or CDC by query; fetches top articles and indexes them

APIs: POST /api/ingest, POST /api/admin/crawl

10. Agent Performance

Layman: Leaderboard — which of the 25 primary agents performs best.

- Comparative line chart (faithfulness, relevancy, precision, recall)
- Top performers list
- Full matrix table by agent

Uses RAGAS data aggregated per agent.

11. Alerts

Layman: Emergency bulletin board for operators.

- Cards by severity: **Critical (red)**, **Warning (yellow)**, **Info**
- Title, message, component, timestamp
- **Resolve** button archives the alert

APIs: GET /api/admin/alerts, PUT /api/admin/alerts/{id}/resolve

12. Smart Routing

Layman: Org chart of how AI hands off work.

- Matrix of **25 agents × 3 tiers**: Primary → Specialist → Human
- Shows trigger rules (e.g. confidence < 70%, frustration > 75)
- Escalation stats summary at top

APIs: GET /api/admin/agent-routing-matrix, GET /api/admin/escalation-stats

13. Escalations

Layman: Safety audit log of handovers.

Metric	Meaning
Conversations processed	Volume baseline
Specialist alerts	Times specialist tier engaged
Human alerts	Times human path triggered
Hot spot	Agent/disease with most escalations

- Filter: All / Specialist / Human
- Timeframe: 7 / 14 / 21 / 30 days
- **Download Excel (CSV)** of escalation log
- Detailed table: agent, disease, names, counts, last event

APIs: GET /api/admin/escalations, GET /api/admin/escalation-stats

14. Quality Metrics (CQS Engine)

Layman: Master scorecard for **Conversation Quality Score (CQS)** — one number summarizing chat excellence.

Six dimensions (weighted):

Dimension	Weight	Examples of what's measured
Engagement	30%	Star ratings, chip clicks, return visits, conversation depth

Response quality	25%	RAGAS faithfulness, relevancy, precision, recall
Clinical safety	20%	Guardrails, escalation accuracy, disclaimers, citations
Session flow	15%	Repeat questions, slot filling, frustration
Format variety	7%	Response format rotation, length, personalization
Velocity	3%	Response speed P50/P95, completion rate

- Click a dimension → see **parameter table** (formula, DB source, weight)
- **Performance matrix** with GREEN / AMBER / RED tiers

API: GET /api/admin/quality/summary

15. Subs & Revenue — *Coming Soon (Sprint 4.0)*

Placeholder for billing, Stripe, and license management.

16. Audit Log — *Coming Soon*

Placeholder for HIPAA-style immutable activity logs.

17. Security & JWT — *Coming Soon*

Placeholder for token rotation analytics and endpoint security logs.

18. Patient Management — *Coming Soon*

Placeholder for EHR sync and longitudinal patient records.

6.4 Admin health check (infrastructure ping)

System Overview and dedicated health logic ping:

- PostgreSQL (latency ms)
- ChromaDB (remote or local fallback)

- LLM provider
- Pre-RAG gatekeeper

API: GET /api/admin/health

7. How Data Flows (Simple Architecture)

Diagram (see online Markdown viewer for full graphic):

```
sequenceDiagram
    participant P as Patient browser
    participant A as Admin browser
    participant API as FastAPI :8000
    participant PG as PostgreSQL
    participant CH as ChromaDB
    participant LLM as ...
```

8. API Reference Summary

Patient-facing

Method	Path	Purpose
POST	/api/auth/token	Login
POST	/api/auth/register	Sign up
GET	/api/auth/me	Profile
POST	/api/chat	Main text chat
POST	/api/chat/image	Image analysis
POST	/api/voice/transcribe	Speech to text
POST	/api/voice/chat	Voice conversation
POST	/api/chat/prescription	PDF prescription
POST	/api/chat/history/download	PDF transcript
GET	/api/history	List conversations
GET	/api/history/search	Search transcripts
POST	/api/feedback	Star rating

GET	/api/conversations/last_active	Resume last chat
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Admin-facing (require admin JWT)

Method	Path	Purpose
GET	/api/admin/overview	Dashboard totals
GET	/api/admin/health	Infrastructure ping
GET	/api/admin/ragas	Quality metrics
GET	/api/admin/sentiment	Sentiment telemetry
GET	/api/admin/feedback	Patient ratings
GET	/api/admin/alerts	Active alerts
PUT	/api/admin/alerts/{id}/resolve	Resolve alert
GET	/api/admin/documents	Indexed docs
GET	/api/admin/vector-store	Chroma collections
GET	/api/admin/prerag/report	Pre-RAG scores
POST	/api/admin/crawl	PubMed/CDC crawl
GET	/api/admin/agent-routing-matrix	Routing tree
GET	/api/admin/escalations	Escalation log
GET	/api/admin/escalation-stats	Escalation KPIs
GET	/api/admin/quality/summary	CQS breakdown
GET	/api/admin/llm-calls	Model call log

Full interactive list: <http://localhost:8000/docs>

9. Operational Tips & Known Limits

Before using the portals

1. Start **PostgreSQL** (`docker compose up -d postgres`).
2. Run `python scripts/setup_db.py` once.
3. Start `python run.py` (backend + both frontends).
4. Ensure `.env` has valid `OPENAI_API_KEY` or `ANTHROPIC_API_KEY` for live chat.

Common issues

Symptom	Likely cause	Fix
"Authorized admin access only"	Logged in as patient, or DB offline dev token	Use <code>admin@prism.ai</code> ; start Postgres; clear browser storage
<code>ECONNREFUSED 127.0.0.1:8000</code>	Backend not running	Run <code>python run.py</code> or <code>uvicorn</code> on port 8000
Empty admin dashboards	DB empty or auth 403	Seed DB; re-login as admin
Chroma "Local fallback"	Remote Chroma container down	<code>docker compose up -d chromadb</code> or use local <code>./data/chroma_db</code>

Clinical disclaimer

PRISM provides **informational AI assistance**. It is **not a substitute** for licensed medical professionals, emergency services, or definitive diagnosis. Human escalation paths exist by design.

Sprint 3.5 scope notes

- **Live:** Patient chat, multimodal, history, feedback, full admin analytics (except Coming Soon modules).
- **Planned (Sprint 4.0):** Revenue, audit log, security dashboard, patient management EHR sync.

Related documents

- [PRISM_Admin_Portal_Documentation.md](#) — Deep dive on admin metrics formulas (Sprint 3.4 reference)
- [SETUP_GUIDE.md](#) — Installation steps for developers

Document generated from PRISM Sprint 3.5 codebase review. For corrections or new modules, update this file when features ship.